University of Utah Spring 2026

MATH 3210-004 PSet 4 Specification

Instructor: Alp Uzman

Subject to Change; Last Updated: 2026-02-12 10:34:50-07:00

1 Background

This problem set is about continuity. The first four problems really are variations of the important theorems real valued continuous functions defined on intervals satisfy, namely the maximum principle (aka extreme value theorem) and the intermediate value theorem.

The final problem is in a sense more general and is about different characterizations of continuity that in different circumstances come in handy. For instance, one characterization of continuity is the vanishing of the oscillation of the function (last part in problem 5), and this characterization is rather useful in the context of problem 3. For the last problem you'll need the notions of open balls, neighborhoods, and nets; here is a quick summary that suffices:

Open Balls and Neighborhoods Let X be a metric space, $x_* \in X$, $r \in \mathbb{R}_{>0}$. Then the **open ball** in X centered at x_* with radius r is by definition the following set:

$$X[x_* \mid < r] = \{x \in X \mid d(x, x_*) < r\}.$$

A **neighborhood** of x_* is a subset $M \subseteq X$ such that M contains some open ball centered at x_*.

Nets (Aka Moore-Smith Sequences) Here is a quick summary of nets, a concept you'll need only for one part of the last problem in this problem set. Recall that for X a set, a sequence in X is nothing but a function of the form $x_\bullet : \mathbb{Z}_{\geq N} \to X$, where $N \in \mathbb{Z}$. Typically we take N to be 0 or 1, but this is not necessary. A net is a generalization of a sequence, in that now one considers more general, possibly multi-dimensional or branching, times. By definition, a **net** (or **generalized sequence** or **Moore-Smith sequence**) in X is a function $x_\bullet : (A, \succcurlyeq) \to X$, where $(A, \succcurlyeq)$ is a **directed set**, that is, a pair where

- A is a set, and
- $\succcurlyeq$ is a relation on A satisfying the following two properties:
  1. For any $\alpha, \beta, \gamma \in A$, if $\alpha \succcurlyeq \beta$ and $\beta \succcurlyeq \gamma$, then $\alpha \succcurlyeq \gamma$.
  2. For any $\alpha_1, \alpha_2 \in A$, there is a $\beta \in A$ such that $\beta \succcurlyeq \alpha_1$ and $\beta \succcurlyeq \alpha_2$.

Here it is instructive to think of elements of A as specific "times", and think of the relation $\alpha \succcurlyeq \beta$ as "α is a time not before β".

We note that $\mathbb{Z}_{\geq N}$ is a directed set for any $N \in \mathbb{Z}$, the ordering being the standard one (that is, $\succcurlyeq = \geq$). One can visualize this directed set associated with sequences like this:

$$N \longrightarrow N+1 \longrightarrow N+2 \longrightarrow N+3 \longrightarrow N+4 \longrightarrow \cdots$$

However not all directed sets are of this form; for instance $(\mathbb{Z}_{\geq 1}, \succcurlyeq)$, where $n \succcurlyeq m$ iff n is divisible by m, too is[1] a directed set, hence can

[1] You should check this!

be used as a "time" for a net. One can visualize this new directed set like so:

$$
\begin{tikzcd}
 & 2 \arrow[r] \arrow[dr] & 4 \arrow[r] & 8 \arrow[r] & \cdots \\
1 \arrow[ur] \arrow[r] \arrow[dr] \arrow[ddr] \arrow[dddr] & 3 \arrow[r] & 6 \arrow[r] & \cdots & \\
 & 5 \arrow[r] & \cdots & & \\
 & 7 \arrow[r] & \cdots & & \\
 & \ddots & & &
\end{tikzcd}
$$

If in addition X is a metric space, one can define convergence of a net in complete analogy with the convergence of a sequence. Indeed, let $x_\infty \in X$ and $x_\bullet : A \to X$ be a net in X. Then $x_\bullet$ is said to **converge** to x_∞ if

$$\forall \varepsilon \in \mathbb{R}_{>0}, \exists \alpha_* \in A, \forall \alpha \in A_{\succcurlyeq \alpha_*} : d(x_\alpha, x_\infty) < \varepsilon,$$

where $A_{\succcurlyeq \alpha_*} = \{\alpha \in A | \alpha \succcurlyeq \alpha_*\}$ is the set of times not before α_*. Thus, in sync with the definition of convergence for sequences, a net $x_\bullet$ converges to x_∞ iff "x_α is within an ε of x_∞ for α later than some time α_*".

2 Generative AI and Computer Algebra Systems Regulations

This section applies only if you choose to use either a generative AI tool (e.g. chatbots) or a computer algebra system (CAS) while working on this problem set. If you do not use such tools, you may skip this section.

The use of these tools while completing this problem set is permitted, provided it is done responsibly and in a manner that supports your learning. Note however that in exams you will not be allowed to use any AI or CAS.

2.1 Disclosure

If you use such tools, you must disclose this fact in the designated section of the form you will complete as part of your submission. You must also share your reflections on which parts of the problem required your own mathematical judgment, insight, or decision-making, and which parts could reasonably be delegated to a computational or generative tool.

2.2 Guidelines for Responsible Use

If you use a generative AI tool or chatbot, you must adhere to the following guidelines:

- During the chat you may share with the chatbot parts or all of this specification document, as well as parts of the textbook or other sources.
- Directly asking the tool for complete problem solutions is prohibited.
- Appropriate uses include asking for conceptual explanations, checking intermediate steps, clarifying definitions or theorems, or exploring alternative solution approaches, provided that the final work submitted is your own.

- Regardless of the tools used, you are expected to understand, justify, and be able to reproduce all submitted work independently. The use of AI or CAS does not reduce or replace this expectation.

3 What to Submit

Submit your detailed solutions to each of the problems below. While the prompts may seem lengthy, the additional text is intended to guide you by providing context and helpful details.

When tackling statements in MATH 3210, follow these four steps (in no particular order):

1. **Assess the truth of the statement:** Determine, either exactly or probabilistically, whether the statement is true.
2. **Build a case:** Come up with examples or begin drafting a proof to identify potential issues or areas where the argument may break down.
3. **Construct a proof:** Provide a formal argument to support your conclusion.
4. **Perturb the statement:** Experiment with logical variations of the statement. Can you prove a stronger/weaker result, or find alternative proofs?

In this problem set, and in later problem sets as well as exams, your work must address the third step (aka constructing a proof). While work regarding the remaining three items likely will help you write a formal proof, understand the material as well as identify connections between different concepts, you do not need to turn in generalizations or multiple proofs of the same statement, nor do you need to provide examples or counterexamples (unless you are specifically asked to). However, demonstrating these steps can contribute to partial credit when appropriate.

Statements in problems will typically be presented in neutral language. For example, a problem may simply state "P" (for P a well-defined statement) rather than "Show that P" or "Prove that P is false". Your task is to determine the truth of the statement and provide a formal proof or refutation.

If a statement is false, it is highly recommended to propose a corrected version of the statement and prove this corrected version. This process, part of "perturbing the statement", will enhance your mathematical error detection and correction skills.

Unless explicitly stated otherwise, all problems require proofs. If you provide an example, you must also prove that the example satisfies the necessary conditions.

Some problems may be (variants of) theorems or exercises from the textbook. Going forward, and possibly for the rest of this problem set, such correspondences will not be mentioned; it'll be up to you to locate them (if need be)!

Make sure that each solution is properly enumerated and organized. Start the solution to each problem on a new page, and consider using headings or subheadings to structure your work clearly. This will not only aid in your thought process but also ensure that no part of your solution is overlooked during grading.

1. Let I be a (nonempty) interval, $f \in F(I;\mathbb{R})$. Then the following are equivalent:

   A: f is continuous.

B: For any $k \in \mathbb{Z}_{\geq 1}$ and any $x_1, ..., x_k \in I$, there is an $x_\dagger \in I$ such that $f(x_\dagger) = \frac{f(x_1)+...+f(x_k)}{k}$.

2. Let $\lambda, \rho \in [-\infty, \infty]$ be with $\lambda < \rho$ and put $I =]\lambda, \rho[$. Then

   (a) There is a continuous function $f : I \to \mathbb{R}$ that is not bounded from above.

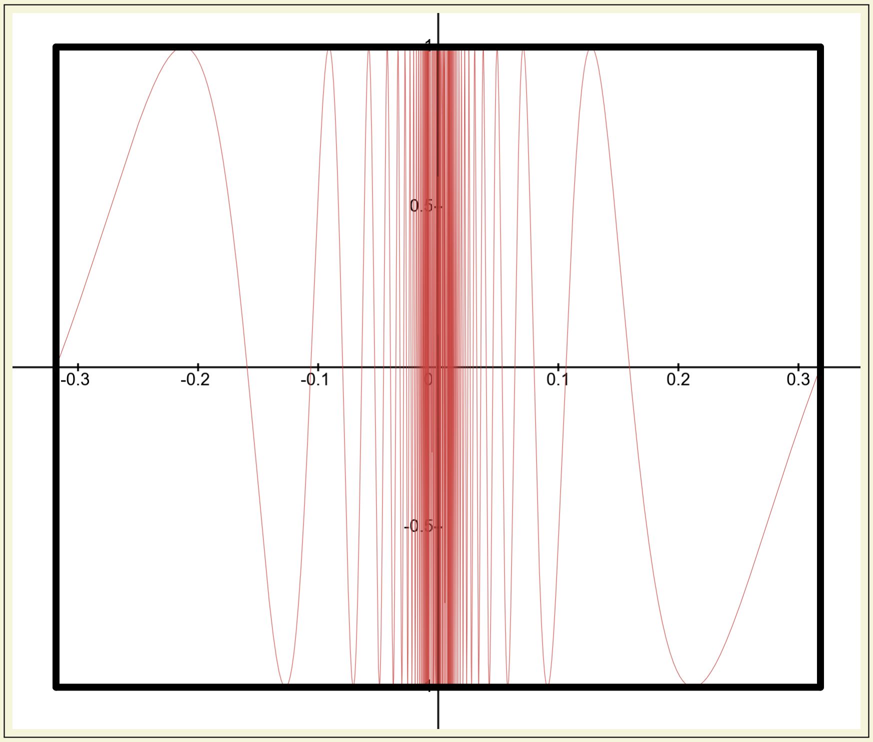
   (b) There is a continuous function $f : I \to \mathbb{R}$ that is bounded from above and whose maximum does not exist.

   It is likely that for either part of this problem, if such a function exist, it will depend on λ and ρ as parameters.

3. Consider the function $f : [-1/\pi, 1/\pi] \to \mathbb{R}$ defined by

$$x \mapsto \begin{cases} \sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0 \end{cases}.$$

   Here is the graph of f:



Further, here is the link to the interactive Desmos graph the above image is generated from:

https://www.desmos.com/calculator/rxxofosueu.

Then

(a) f is continuous.

(b) f takes any value between any two values it takes. (Less cryptically, it satisfies the conclusion of the Intermediate Value Theorem.)

(c) The function $g : [-1/\pi, 1/\pi] \to \mathbb{R}$ defined by $x \mapsto xf(x)$ is continuous.

(d) The function $g : [-1/\pi, 1/\pi] \to \mathbb{R}$ defined by $x \mapsto xf(x)$ is takes any value between any two values it takes.

(e) If $h : [-1/\pi, 1/\pi] \to \mathbb{R}$ is a function takes any value between any two values it takes, and it takes any value exactly once, then h is continuous.

(f) If $h : [-1/\pi, 1/\pi] \to \mathbb{R}$ is a function takes any value between any two values it takes, and it takes any value at most finitely many times, then h is continuous.

4. **For this problem, if you so prefer you may assume $X = \mathbb{R}$.** Let X be a metric space and $A \subseteq X$. Then the following are equivalent:

(a) A is compact (that is, any sequence in A has a subsequence that converges to some point in A).

(b) For any $f \in F(A; \mathbb{R})$, if f is continuous, then it is bounded.

(c) For any $f \in F(A; \mathbb{R})$, if f is continuous, then im(f) is compact.

(d) For any $f \in F(A;\mathbb{R})$, if f is continuous and bounded, then max(f) and min(f) exist.

5. **For this problem, if you so prefer you may assume** $X, Y \subseteq \mathbb{R}$**.** Let $f : X \to Y$ be a function between metric spaces, $x_* \in X$. Then the following are equivalent:

   (a) f is continuous at x_*.

   (b) For any net $x_\bullet$ that converges to x_*, $f(x_\bullet)$ is a net that converges to $f(x_*)$.

   (c) For any $\varepsilon \in \mathbb{R}_{>0}$, there is a $\delta \in \mathbb{R}_{>0}$ such that $f(X[x_* \mid < \delta]) \subseteq Y[f(x_*) \mid < \varepsilon]$.

   (d) For any neighborhood $N \subseteq Y$ of $f(x_*)$, there is a neighborhood $M \subseteq X$ of x_* such that $f(M) \subseteq N$.

   (e) The oscillation of f at x_* is zero, where the **oscillation** is defined by

$$\operatorname{osc}(f; x_*) = \inf\left\{ \operatorname{diam}(f(X[x_* \mid < \delta])) \,\middle|\, \delta \in \mathbb{R}_{>0} \right\},$$

   where the diameter of a subset $B \subseteq Y$ is by definition $\operatorname{diam}(B) = \sup_{y_1, y_2 \in Y} d_Y(y_1, y_2) \in [0, \infty]$.

4 How to Submit

- **Step 1 of 2:** Submit the form at the following URL:

  https://forms.gle/iNY3sJURXonQqQGX6.

  You will receive a zero for this assignment if you skip this step, even if you submit your work on Gradescope on time.

- **Step 2 of 2:** Submit your work on Gradescope at the following URL:

  https://www.gradescope.com/courses/1212058/assignments/7390761,

  see the Gradescope documentation for instructions.

5 When to Submit

This problem set is due on February 13, 2026 at 11:59 PM.

As per the course's syllabus, late submissions, up to 24 hours after this deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before the 24-hour extension expires.